

解三角形

一、在 $\triangle ABC$ 中， $\angle A, \angle B, \angle C$ 所对边分别为 a, b, c .

(1) 已知 $a=2, A=\frac{\pi}{3}$, 求 $\triangle ABC$ 周长和面积范围

变式: 若 $\triangle ABC$ 为锐角三角形, 且其余条件不变 (问题同上)

变式2: 条件同变式1, 求 $\triangle ABC$ 内切圆半径范围

(2) 已知 $b=2, A=\frac{\pi}{3}$, 求 $\triangle ABC$ 周长和面积范围

变式1: 若 $\triangle ABC$ 为锐角三角形, 且其余条件不变 (问题同上)

(3) 已知 $a=2, B=C$, 且 $\triangle ABC$ 为锐角三角形, 求 $\triangle ABC$ 周长和面积和内切圆半径范围

(4) 已知 $a=2, B=2C$, 且 $\triangle ABC$ 为锐角三角形, 求 $\triangle ABC$ 周长和面积范围

变式: 若将 $B=2C$ 改成 $A=2C$, 其余同上, 问题不变

(5) 已知 $A=\frac{\pi}{3}, a=2$, D 是 BC 上中点, 求 AD 最大值

变式: 若 D 改为 BC 上靠近 C 的三等分点, 其余同上, 问题同样

(6) 已知 $A=\frac{\pi}{3}, b+c=2$, AD 是 $\angle BAC$ 的角平分线, 求 AD 最大值

变式: 若 $b+c=2$ 改为 $a=2$, 其余同上, 问题同样

(7) 已知 $C=B+\frac{\pi}{2}$, 求 $\frac{a+b}{c^2}$ 最小值, 变式: 问题改成求 $3\tan C + \tan A$ 最小值 (其余不变)

(8) 已知 $\angle ABC=\frac{2\pi}{3}$, D 是 AC 上一点, 且 $\angle DBC=\frac{\pi}{2}$, $BD=3$, 求 $a+2c$ 最小值

(9) 已知 $2\cos C = \frac{2b}{a} - \frac{a}{b}$, $\triangle ABC$ 为锐角三角形, 求 B 范围

二、判断下列 $\triangle ABC$ 形状 (在 $\triangle ABC$ 中, 角 A, B, C 所对边分别为 a, b, c)

(1) $b-a=2b\sin^2\frac{C}{2}$

(2) $a\cos A + b\cos B = c\cos C$

(3) $(a^2+b^2)\sin(A-B) = (a^2-b^2)\sin C$

(4) D 是 BC 边上中点, 且 $\angle BAD + \angle ACB = 90^\circ$

(5) $a\sin A = b\cos C + c\cos B$

(6) $\frac{a}{\cos A} = \frac{b}{\cos B} = \frac{c}{\cos C}$

从“等腰”三角形, “直角”三角形, “等腰直角”三角形, “等腰或直角”三角形; “等边”三角形中选)

三、完成下列的求值

(1) 已知 $a=1, A=\frac{\pi}{6}, c=\sqrt{3}$, 求角 C 值

(2) 已知 $a=2, b+c=3, bc=2$, 求 $\cos A$ 值

(3) 已知 $c=1, a\cos B = 2\sin C, \cos A \sin B = \frac{1}{8}$, 求 $\sin C$ 值

(4) 已知 $A+B=3C, \sin B = 2\sin(A-C)$, 求 $\sin A$ 值

(5) 已知 $a\cos B = \frac{3}{5}, b\cos A = \frac{2}{5}$, 求 $\frac{\sin(A-B)}{\sin C}$ 值

(6) 已知 $\sin(B-C)\tan A = \sin B \sin C$, 求 $\frac{a^2+b^2}{c^2}$ 值

(7) 已知 $a^2+b^2=3c^2$, 求 $\frac{1}{\tan A} + \frac{1}{\tan B} - \frac{1}{\tan C}$ 值

(8) 已知 $a+c=2b$, 求 $\tan \frac{A}{2} \tan \frac{C}{2}$ 值

(9) 已知 $a\cos B - b\cos A = \frac{3}{5}c$, 求 $\frac{\tan A}{\tan B}$ 值

(10) 已知 $\frac{b}{a} + \frac{a}{b} = 6\cos C$, 求 $\frac{\tan C}{\tan A} + \frac{\tan C}{\tan B}$ 值

(11) 已知 $C=\frac{\pi}{3}, \sqrt{a}=3c, \triangle ABC$ 是锐角三角形, 求 $\frac{1}{\tan A} + \frac{1}{\tan B}$ 值

(12) 已知 $c=2b\cos B, C=\frac{2\pi}{3}$, 求角 B 值

(13) 已知 $a\sin \frac{A+C}{2} = b\sin A$, 求角 B 值

四. 综合题

1. 已知 D 是 AB 边上靠近点 B 的三等分点, 且 $CD = 2$, $\angle ACB = 120^\circ$, $AB = 6$, 求 $\triangle ABC$ 面积

2. 已知 $\frac{a - \cos B}{a - \cos C} = \frac{b}{c}$, 且 $b \neq c$, 求证: $a^2 = b + c$

3. 已知 D 是 AC 边上一点, $BD \sin \angle ABC = a \sin C$, $AD = 2CD$, 且 $b^2 = ac$, 求 $\cos \angle ABC$ 值

4. 已知 $a^2 = b^2 + bc$, 求证: $A = 2B$

5. 已知 $\frac{\cos A}{1 + \sin A} = \frac{\sin 2B}{1 + \cos 2B}$, 求证: $C = \frac{\pi}{2} + B$

6. 已知 $\frac{b^2}{a^2} = 1 + \sin 2C$, $0 < C < \frac{3}{4}\pi$, 求角 A 值

7. 已知 $A = \frac{\pi}{4}$, $ac = 2\sqrt{10}$, $\frac{\sin C}{\sin B} = \frac{2\sqrt{2}}{3}$, 求 $\triangle ABC$ 面积

8. 已知平面四边形 $ABCD$ 中, $\angle A = 135^\circ$, $\angle B = \angle D = 90^\circ$, $AB = \sqrt{2}AD = 2$, 求四边形 $ABCD$ 面积

9. 已知 $c^2 \sin A = b \sin C$, $(a+c)^2 = 18 + b^2$, 求 $\triangle ABC$ 面积

10. 已知平面四边形 $ABCD$ 内接于圆 O , $AB = CD = 5$, $AD = 3$, $\angle BCD = 60^\circ$, 求四边形 $ABCD$ 面积

11. 已知梯形 $ABCD$ 中, $AD \parallel BC$, 设 $\angle BAD = \alpha$, $\angle ABD = \beta$, $\cos(\alpha - \beta) = 2\sin(\alpha + \frac{\pi}{3})\sin(\beta + \frac{\pi}{3})$, 求 $\angle ADB$

12. 已知 $\sin A \sin B + \sin B \sin C + \cos 2B = 1$, $C = \frac{3}{4}\pi$, 求 $\frac{\sin A}{\sin B}$ 值

13. 已知平面四边形 $ABCD$ 中, $AD = 2AB = 4$, $\angle DAB + \angle DCB = \pi$, AC 是 $\angle DAB$ 的角平分线, 求 $\frac{BD^4}{BC \cdot CD}$ 最大值

14. 已知 D 是线段 AC 上一点, $\vec{AD} = 2\vec{DC}$, $AC = 3BC$, $\sin \angle BDC = 3 \sin \angle BAC$, 当 $\vec{CA} \cdot \vec{CB} - |\vec{AB}|$ 最小时, 求此时 $\triangle ABC$ 面积

15. 已知平面四边形 $ABCD$ 中, $AB \perp AD$, $AB = 1$, $AD = \sqrt{3}$, $BC = \sqrt{2}$

(1) 若 $CD = 2$, 求 $\sin \angle ADC$ (2) 若 $\angle C = \frac{\pi}{4}$, 求四边形 $ABCD$ 面积

16. 已知平面四边形 $ABCD$ 中, $AB \perp AD$, $\angle ACD = \frac{\pi}{3}$, $AD = \sqrt{3}$, $2S = -\sqrt{3} \vec{BA} \cdot \vec{BC}$ (其中 S 是 $\triangle ABC$ 面积) (1) 求角 B (2) 若 $\cos \angle ADC = \frac{1}{2}$, 求四边形 $ABCD$ 周长

17. 已知 $\triangle ABC$ 中, D 是 BC 边上一点, 且 $BD = AC$, $\angle BCA = 40^\circ$, $\angle CAD = 60^\circ$, 求 $\angle ABD$

18. 已知 $\triangle ABC$ 中, D 是 BC 上一点, $BA \perp AD$, 且 $\angle BAC = 120^\circ$, 由 $AD = 2$ 求 $\triangle ABC$ 面积最小值

19. 已知在 $\triangle ABC$ 中, $AB = 2\sqrt{2}$, $BC = 4$, $AC = 2\sqrt{10}$; M 是线段 BC 上中点, P 是线段 AB 延长线上一点, 求 $\angle CPM$ 最大值

解三角形.

一. (1) 由 $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$, 则 $b = \frac{4}{\sqrt{3}} \sin B$, $c = \frac{4}{\sqrt{3}} \sin C$, 又由 $a=2$, $B=\pi-A-C$,

则周长 $C_{\triangle ABC} = a+b+c = 2 + \frac{4}{\sqrt{3}} (\sin B + \sin C) = 2 + \frac{4}{\sqrt{3}} (\sin(\frac{2}{3}\pi - C) + \sin C)$

$= 2 + \frac{4}{\sqrt{3}} (\frac{\sqrt{3}}{2} \cos C + \frac{1}{2} \sin C) = 2 + 2(\cos C + \sqrt{3} \sin C) = 2 + 4 \sin(C + \frac{\pi}{6})$

由 $C \in (0, \frac{2\pi}{3})$, 则 $C + \frac{\pi}{6} \in (\frac{\pi}{6}, \frac{5\pi}{6})$ 则 $C_{\triangle ABC} \in (4, 6]$ (最大值在 $C = \frac{\pi}{3}$ 时取得)

则面积 $S_{\triangle ABC} = \frac{1}{2} ab \sin C = \frac{4}{\sqrt{3}} \sin B \sin C = \frac{4}{\sqrt{3}} \sin(\frac{2}{3}\pi - C) \sin C = \frac{4}{\sqrt{3}} (\frac{\sqrt{3}}{2} \cos C + \frac{1}{2} \sin C) \sin C$

$= \frac{2}{\sqrt{3}} (\sqrt{3} \cos C \sin C + \sin^2 C) = \frac{2}{\sqrt{3}} (\frac{\sqrt{3}}{2} \sin 2C + \frac{1 - \cos 2C}{2}) = \frac{2}{\sqrt{3}} (\frac{\sqrt{3}}{2} \sin 2C - \frac{1}{2} \cos 2C + \frac{1}{2})$

$= \frac{2}{\sqrt{3}} \sin(2C - \frac{\pi}{6}) + \frac{1}{\sqrt{3}}$ 由 $C \in (0, \frac{2}{3}\pi)$ 则 $2C - \frac{\pi}{6} \in (-\frac{\pi}{6}, \frac{7}{6}\pi)$

则 $S_{\triangle ABC} \in (0, \sqrt{3}]$ (最大值在 $C = \frac{\pi}{3}$ 时取得)

变式1: 由 $\triangle ABC$ 为锐角三角形, 则 $\begin{cases} 0 < B < \frac{\pi}{2} \\ 0 < C < \frac{\pi}{2} \end{cases}$ 又由 $B = \frac{2}{3}\pi - C$, 则 $0 < \frac{2}{3}\pi - C < \frac{\pi}{2}$

即 $C \in (\frac{\pi}{6}, \frac{\pi}{2})$ 则 $C + \frac{\pi}{6} \in (\frac{\pi}{3}, \frac{5}{6}\pi)$, 则 $C_{\triangle ABC} \in (2\sqrt{3} + 2, 6]$

又由 $2C - \frac{\pi}{6} \in (\frac{\pi}{6}, \frac{5}{6}\pi)$, 则 $S_{\triangle ABC} \in (\frac{2\sqrt{3}}{3}, \sqrt{3}]$

变式2: 设 $\triangle ABC$ 内切圆半径为 r , 则 $r = \frac{2S_{\triangle ABC}}{C_{\triangle ABC}} = \frac{\frac{2}{\sqrt{3}} [2 \sin(2C - \frac{\pi}{6}) + 1]}{2 + 4 \sin(C + \frac{\pi}{6})}$

$= \frac{1}{\sqrt{3}} \cdot \frac{2 \sin(2C - \frac{\pi}{6}) + 1}{1 + 2 \sin(C + \frac{\pi}{6})}$ 设 $C + \frac{\pi}{6} = \theta$ ($\theta \in (\frac{\pi}{3}, \frac{5}{6}\pi)$), 则 $r = \frac{1}{\sqrt{3}} \cdot \frac{-2 \cos 2\theta - 1}{1 + 2 \sin \theta} = -\frac{1}{\sqrt{3}} \cdot \frac{2 \cos 2\theta + 1}{1 + 2 \sin \theta}$

设 $f(\theta) = \frac{2 \cos 2\theta + 1}{1 + 2 \sin \theta}$, 则 $f'(\theta) = \frac{-2 \cos \theta (4 \sin^2 \theta + 4 \sin \theta + 3)}{(1 + 2 \sin \theta)^2}$ 由 $4 \sin^2 \theta + 4 \sin \theta + 3 > 0$.

令 $f'(\theta) < 0$, 则 $\theta \in (\frac{\pi}{3}, \frac{\pi}{2})$ 时, $f(\theta)$ 单调递减, r 单调递增.

令 $f'(\theta) > 0$, 则 $\theta \in (\frac{\pi}{2}, \frac{5}{6}\pi)$ 时, $f(\theta)$ 单调递增, r 单调递减.

则 $r \in (0, \frac{\sqrt{3}}{9}]$

(2) 由 $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$, 则 $c = \frac{b \sin C}{\sin B} = \frac{2 \sin C}{\sin B}$, $a = \frac{b \sin A}{\sin B} = \frac{\sqrt{3}}{\sin B}$

则 $C_{\triangle ABC} = a+b+c = \frac{\sqrt{3} + 2 \sin C}{\sin B} + 2 = \frac{\sqrt{3} + 2 \sin(\frac{2}{3}\pi - B)}{\sin B} + 2 = \frac{\sqrt{3} + \sqrt{3} \cos B + \sin B}{\sin B} + 2$

$= \frac{\sqrt{3}(1 + \cos B)}{\sin B} + 3 = \frac{2\sqrt{3} \cos \frac{B}{2}}{2 \sin \frac{B}{2} \cos \frac{B}{2}} + 3 = \frac{\sqrt{3}}{\tan \frac{B}{2}} + 3$ 由 $B \in (0, \frac{2}{3}\pi)$ 则 $\frac{B}{2} \in (0, \frac{\pi}{3})$ 则 $C_{\triangle ABC} \in (4, +\infty)$

则 $S_{\triangle ABC} = \frac{1}{2} bc \sin A = \frac{\sqrt{3}}{2} \cdot \frac{2 \sin C}{\sin B} = \sqrt{3} \frac{\sin(\frac{2}{3}\pi - B)}{\sin B} = \sqrt{3} \cdot \frac{\frac{\sqrt{3}}{2} \cos B + \frac{1}{2} \sin B}{\sin B} = \frac{3}{2 \tan B} + \frac{\sqrt{3}}{2}$

则 $S_{\triangle ABC} \in (0, +\infty)$

变式1: 由 $\triangle ABC$ 为锐角三角形. 则 $\begin{cases} 0 < B < \frac{\pi}{2} \\ 0 < \frac{2}{3}\pi - B < \frac{\pi}{2} \end{cases}$ 即 $B \in (\frac{\pi}{6}, \frac{\pi}{2})$

则 $\frac{B}{2} \in (\frac{\pi}{12}, \frac{\pi}{4})$, 由 $\tan \frac{\pi}{12} = \tan(\frac{\pi}{3} - \frac{\pi}{4}) = \frac{\sqrt{3} - 1}{1 + \sqrt{3} \cdot 1} = 2 - \sqrt{3}$, 即 $C_{\triangle ABC} \in (3 + \sqrt{3}, 6 + 2\sqrt{3})$

则 $B \in (\frac{\pi}{6}, \frac{\pi}{2})$ 即 $S_{\triangle ABC} \in (\frac{\sqrt{3}}{2}, 2\sqrt{3})$

(3) 由 $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$, 则由 $B=C$, 则 $\frac{2}{\sin A} = \frac{b}{\sin(\frac{\pi}{2} - \frac{A}{2})} = \frac{c}{\sin(\frac{\pi}{2} - \frac{A}{2})}$

即 $b=c = \frac{2 \cos \frac{A}{2}}{\sin A} = \frac{1}{\sin \frac{A}{2}}$, 则 $C_{\triangle ABC} = a+b+c = 2 + \frac{2}{\sin \frac{A}{2}}$ 又由 $\begin{cases} 0 < \frac{\pi}{2} - \frac{A}{2} < \frac{\pi}{2} \\ 0 < A < \frac{\pi}{2} \end{cases}$

则 $0 < A < \frac{\pi}{2}$ 即 $\frac{A}{2} \in (0, \frac{\pi}{4})$ 即 $C_{\triangle ABC} \in (2 + 2\sqrt{2}, +\infty)$

则 $S_{\triangle ABC} = \frac{1}{2} bc \sin A = \frac{1}{2} \cdot \frac{\sin A}{\sin^2 \frac{A}{2}} = \frac{1}{2} \cdot \frac{2 \sin \frac{A}{2} \cos \frac{A}{2}}{\sin^2 \frac{A}{2}} = \frac{1}{\tan \frac{A}{2}}$, 即 $S_{\triangle ABC} \in (1, +\infty)$

由 $r = \frac{2S_{\triangle ABC}}{C_{\triangle ABC}} = \frac{2 \cdot \frac{1}{\tan \frac{A}{2}}}{2(1 + \frac{1}{\sin \frac{A}{2}})} = \frac{\cos \frac{A}{2}}{\sin \frac{A}{2} + 1} = \frac{\cos^2 \frac{A}{4} - \sin^2 \frac{A}{4}}{(\sin \frac{A}{4} + \cos \frac{A}{4})^2} = \frac{\cos \frac{A}{4} - \sin \frac{A}{4}}{\sin \frac{A}{4} + \cos \frac{A}{4}} = \frac{1 - \tan \frac{A}{4}}{\tan \frac{A}{4} + 1} = -1 + \frac{2}{\tan \frac{A}{4} + 1}$

由 $A \in (0, \frac{\pi}{2})$ 则 $\frac{A}{4} \in (0, \frac{\pi}{8})$ 由 $\tan \frac{\pi}{8} = \frac{1 - \cos \frac{\pi}{4}}{\sin \frac{\pi}{4}} = \sqrt{2} - 1$, 则 $r \in (\sqrt{2} - 1, 1)$

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(4). 由 $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$, 则 $\frac{2}{\sin(\pi-3C)} = \frac{b}{\sin 2C} = \frac{c}{\sin C}$, 则 $c = \frac{2 \sin C}{\sin(\pi-3C)} = \frac{2 \sin C}{\sin 3C}$
 $= \frac{2 \sin C}{\sin(2C+C)} = \frac{2 \sin C}{\sin 2C \cos C + \cos 2C \sin C} = \frac{2 \sin C}{2 \sin C \cos^2 C + (2 \cos^2 C - 1) \sin C} = \frac{2}{4 \cos^2 C - 1}$ 则 $b = \frac{c \sin 2C}{\sin C} = 2 \cos C \cdot c$
 $= \frac{4 \cos C}{4 \cos^2 C - 1}$ 由 $\triangle ABC = a + b + c = 2 + \frac{4 \cos C + 2}{4 \cos^2 C - 1}$ 由 $\begin{cases} 0 < C < \frac{\pi}{2} \\ 0 < B < \frac{\pi}{2} \\ 0 < A < \frac{\pi}{2} \end{cases}$ 则 $\begin{cases} 0 < C < \frac{\pi}{2} \\ 0 < 2C < \frac{\pi}{2} \\ 0 < \pi - 3C < \frac{\pi}{2} \end{cases}$
 即 $C \in (\frac{\pi}{6}, \frac{\pi}{4})$ 则 $\cos C \in (\frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2})$ 设 $4 \cos C + 2 = t$ ($t \in (2\sqrt{2}+2, 2\sqrt{3}+2)$)

则 $C_{\triangle ABC} = 2 + \frac{t}{4(\frac{t-2}{4})^2 - 1} = 2 + \frac{4}{t-4}$ 则 $C_{\triangle ABC} \in (3+\sqrt{3}, 4+2\sqrt{2})$

由 $S_{\triangle ABC} = \frac{1}{2} ab \sin C = \frac{4 \cos C \sin C}{4 \cos^2 C - 1} = \frac{2 \sin 2C}{2 \cos 2C + 1}$ 设 $2C = \theta$ ($\theta \in (\frac{\pi}{3}, \frac{\pi}{2})$)

且设 $f(\theta) = \frac{2 \sin \theta}{2 \cos \theta + 1}$, 则 $f'(\theta) = \frac{2(\cos \theta + 2)}{(2 \cos \theta + 1)^2} > 0$. 则 $f(\theta)$ 单调递增.

即 $S_{\triangle ABC} \in (\frac{\sqrt{3}}{2}, 2)$

变式: 由 $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$, $a=2$, $A=2C$, 则 $\frac{2}{\sin 2C} = \frac{b}{\sin(\pi-3C)} = \frac{c}{\sin C}$

则 $c = \frac{2 \sin C}{\sin 2C} = \frac{1}{\cos C}$, $b = \frac{c \sin(\pi-3C)}{\sin C} = \frac{c \sin 3C}{\sin C} = \frac{c \sin C \cos 2C + \sin 2C \cos C}{\sin C} = (4 \cos^2 C - 1) \cdot c$
 $= \frac{4 \cos^2 C - 1}{\cos C} = 4 \cos C - \frac{1}{\cos C}$ 由 $\begin{cases} 0 < A < \frac{\pi}{2} \\ 0 < B < \frac{\pi}{2} \\ 0 < C < \frac{\pi}{2} \end{cases}$ 则 $\begin{cases} 0 < 2C < \frac{\pi}{2} \\ 0 < \pi - 3C < \frac{\pi}{2} \\ 0 < C < \frac{\pi}{2} \end{cases}$ 即 $C \in (\frac{\pi}{6}, \frac{\pi}{4})$

则 $C_{\triangle ABC} = a + b + c = 2 + 4 \cos C$, 则 $C_{\triangle ABC} \in (2\sqrt{2}+2, 2\sqrt{3}+2)$

则 $S_{\triangle ABC} = \frac{1}{2} ab \sin C = 4 \cos C \cdot \sin C - \frac{\sin C}{\cos C} = 2 \sin 2C - \tan C$, 设 $C = \theta$ ($\theta \in (\frac{\pi}{6}, \frac{\pi}{4})$)

设 $f(\theta) = 2 \sin 2\theta - \tan \theta$, 则 $f'(\theta) = 4 \cos 2\theta - \frac{1}{\cos^2 \theta} = \frac{8 \cos^4 \theta - 4 \cos^2 \theta - 1}{\cos^2 \theta}$

令 $f'(\theta) = 0$, 则 $\cos^2 \theta = \frac{1-\sqrt{3}}{4}$ (舍) 或 $\frac{1+\sqrt{3}}{4}$ 即 $\cos \theta = \frac{\sqrt{1+\sqrt{3}}}{2}$ 或 $-\frac{\sqrt{1+\sqrt{3}}}{2}$ (舍)

由 $\theta \in (\frac{\pi}{6}, \frac{\pi}{4})$, 则 $\cos \theta \in (\frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2})$ 当 $\cos \theta \in (\frac{\sqrt{2}}{2}, \frac{\sqrt{1+\sqrt{3}}}{2})$ 时, $f(\theta)$ 单调递减,

当 $\cos \theta \in (\frac{\sqrt{1+\sqrt{3}}}{2}, \frac{\sqrt{3}}{2})$ 时, $f(\theta)$ 单调递增. 则 $S_{\triangle ABC} \in [\sqrt{2}-\sqrt{3}, 1)$.

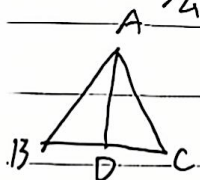
(5.1) : 法一: 向量法

由 $\vec{AD} = \frac{1}{2} \vec{AB} + \frac{1}{2} \vec{AC}$, 则 $AD^2 = \frac{1}{4} (c^2 + b^2 + bc)$ 又由余弦定理: $a^2 = b^2 + c^2 - 2bc \cos \frac{A}{2} = 4$

则 $b^2 + c^2 - bc = 4$ 则 $AD^2 = \frac{1}{4} (4 + 2bc) = \frac{2+bc}{2}$ 由 $4 + bc = b^2 + c^2 \geq 2bc$ (当 $b=c$ 时取等)

即 $bc \leq 4$ 则 $AD^2 = \frac{2+bc}{2} \leq 3$ 即 $|AD| \leq \sqrt{3}$ 则 AD 最大值为 $\sqrt{3}$.

法二: 双余弦



由 $\cos \angle BDA = \frac{AD^2 + BD^2 - AB^2}{2AD \cdot BD} = \frac{AD^2 + 1 - c^2}{2AD}$, $\cos \angle ADC = \frac{AD^2 + CD^2 - AC^2}{2AD \cdot CD} = \frac{AD^2 + 1 - b^2}{2AD}$

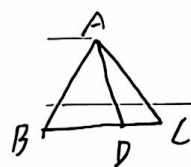
又由 $\angle BDA + \angle ADC = \pi$ 则 $\cos \angle BDA + \cos \angle ADC = \frac{2AD^2 + 2 - c^2 - b^2}{2AD} = 0$

即 $AD^2 = \frac{c^2 + b^2 - 2}{2} = \frac{c^2 + b^2}{2} - 1$ 由余弦定理: $b^2 + c^2 - bc = 4$

则 $AD^2 = \frac{4+bc}{2} - 1 = \frac{bc}{2} + 1$ 又由 $4 + bc = b^2 + c^2 \geq 2bc$ (当且仅当 $b=c$ 时取等)

即 $bc \leq 4$ 即 $AD^2 = \frac{bc}{2} + 1 \leq 3$ 则 $AD \leq \sqrt{3}$ 即 AD 最大值为 $\sqrt{3}$.

5. 变式：法一：向量法：



由 $\vec{AD} = \frac{2}{3}\vec{AC} + \frac{1}{3}\vec{AB}$ ，则 $AD^2 = \frac{4AC^2 + AB^2 + 4\vec{AC} \cdot \vec{AB}}{9} = \frac{4b^2 + c^2 + 2bc}{9}$

又由余弦定理： $a^2 = b^2 + c^2 - 2bc \cdot \frac{1}{2} = 2^2$ 则 $b^2 + c^2 - bc = 4$ 则 $AD^2 = \frac{6b^2 + 3c^2 - 8}{9}$

由 $b^2 + c^2 = 4 + bc = 4 + \frac{1}{\lambda}(2b) \cdot c \leq 4 + \frac{1}{\lambda} \frac{(2b)^2 + c^2}{2} = 4 + \frac{2b^2}{\lambda} + \frac{c^2}{2\lambda}$ (其中 $\lambda > 0$)

由 $b^2 + c^2 \leq 4 + \frac{2}{\lambda}b^2 + \frac{1}{2\lambda}c^2$ 则 $b^2(1 - \frac{2}{\lambda}) + c^2(1 - \frac{1}{2\lambda}) \leq 4$ 令 $\frac{1 - \frac{2}{\lambda}}{1 - \frac{1}{2\lambda}} = \frac{2}{1}$ ，则

解得 $\lambda = \sqrt{3} - 1$ 或 $-\sqrt{3} - 1$ (舍) 则有 $\frac{3 - \sqrt{3}}{4}(2b^2 + c^2) \leq 4$ 即 $2b^2 + c^2 \leq \frac{8(3 + \sqrt{3})}{3}$

则 $AD^2 = \frac{6b^2 + 3c^2 - 8}{9} = \frac{3(2b^2 + c^2) - 8}{9} \leq \frac{16 + 8\sqrt{3}}{9}$ (当且仅当 $c = (\sqrt{3} - 1)b$ ，即 $c = (\sqrt{3} - 1)b = \frac{2\sqrt{6}}{3}$ 取等)

则 $|AB| \leq \frac{2}{3}(\sqrt{3} + 1) = \frac{2\sqrt{3} + 2}{3}$ 即 AD 最大值为 $\frac{2\sqrt{3} + 2}{3}$

法二：双余弦：

由 $\cos \angle BDC = \frac{BD^2 + AD^2 - AB^2}{2BD \cdot AD} = \frac{(\frac{2}{3})^2 + AD^2 - c^2}{2 \cdot \frac{2}{3} \cdot AD} = \frac{\frac{4}{9} + AD^2 - c^2}{\frac{4}{3}AD}$ ， $\cos \angle ADC = \frac{CD^2 + AD^2 - AC^2}{2CD \cdot AD} = \frac{(\frac{1}{3})^2 + AD^2 - b^2}{2 \cdot \frac{1}{3} \cdot AD} = \frac{\frac{1}{9} + AD^2 - b^2}{\frac{2}{3}AD}$

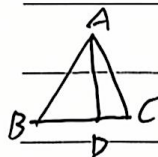
又由 $\angle BDC + \angle ADC = \pi$ ，则 $\cos \angle BDC + \cos \angle ADC = \frac{\frac{4}{9} + AD^2 - c^2}{\frac{4}{3}AD} + \frac{\frac{1}{9} + AD^2 - b^2}{\frac{2}{3}AD} = \frac{\frac{2}{9} + 3AD^2 - 2b^2 - c^2}{\frac{8}{3}AD} = 0$

即 $AD^2 = \frac{2b^2 + c^2 - \frac{2}{9}}{3} = \frac{2b^2 + c^2}{3} - \frac{8}{9}$ 由 $b^2 + c^2 = 4 + bc = 4 + \frac{1}{\sqrt{3}-1}(\sqrt{3}-1)bc \leq 4 + \frac{1}{\sqrt{3}-1} \frac{[(\sqrt{3}-1)b]^2 + c^2}{2}$

则 $\frac{3 - \sqrt{3}}{4}(2b^2 + c^2) \leq 4$ 即 $2b^2 + c^2 \leq \frac{8(3 + \sqrt{3})}{3}$ (当且仅当 $c = (\sqrt{3} - 1)b = \frac{2\sqrt{6}}{3}$ 时取等)

则 $AD^2 = \frac{8}{9}(2 + \sqrt{3}) = \frac{4}{9}(4 + 2\sqrt{3}) = \frac{4}{9}(\sqrt{3} + 1)^2$ ，即 $AD \leq \frac{2(\sqrt{3} + 1)}{3} = \frac{2\sqrt{3} + 2}{3}$

6. 由 AD 是角平分线，则 $\angle BAD = \angle CAD = \frac{\pi}{6}$ 则 $S_{\triangle ABC} = S_{\triangle ABD} + S_{\triangle ADC}$



即 $\frac{1}{2} \cdot bc \sin \frac{\pi}{3} = \frac{1}{2} \cdot c \cdot AD \cdot \sin \frac{\pi}{6} + \frac{1}{2} \cdot b \cdot AD \sin \frac{\pi}{6}$ 则 $AD = \frac{\sqrt{3}bc}{b+c}$

由 $b+c=2$ ，则 $bc \leq (\frac{b+c}{2})^2 = 1$ (当且仅当 $b=c$ 时取等) 则 $AD = \frac{\sqrt{3}bc}{b+c} = \frac{\sqrt{3}bc}{2} \leq \frac{\sqrt{3}}{2}$

即 AD 最大值为 $\frac{\sqrt{3}}{2}$ (当且仅当 $b=c=1$ 时取等)

变式：由原式，则 $AD = \frac{\sqrt{3}bc}{b+c}$ ，又由余弦定理 $a^2 = b^2 + c^2 - 2bc \cdot \frac{1}{2} = 2^2$ 即 $b^2 + c^2 - bc = 4$

由 $(b+c)^2 - 3bc = 4$ 即 $(b+c)^2 = 4 + 3bc \leq 4 + 3(\frac{b+c}{2})^2$ 则 $\frac{(b+c)^2}{4} \leq 4$ 即 $b+c \leq 4$ (当且仅当 $b=c=2$ 时取得等号) 则 $AD = \frac{\sqrt{3}bc}{b+c} \leq \frac{\sqrt{3}(\frac{b+c}{2})^2}{b+c} = \frac{\sqrt{3}}{4}(b+c) \leq \sqrt{3}$ (当且仅当 $b=c=2$ 时取等)

则 AD 最大值为 $\sqrt{3}$ 。

7. 由 $C = B + \frac{\pi}{2}$ ，则 $B = C - \frac{\pi}{2}$ ， $A = \pi - B - C = \pi - (C - \frac{\pi}{2}) - C = \frac{3}{2}\pi - 2C$

由 $\frac{a^2 + b^2}{c^2} = \frac{\sin^2 A + \sin^2 B}{\sin^2 C} = \frac{\sin^2(\frac{3}{2}\pi - 2C) + \sin^2(C - \frac{\pi}{2})}{\sin^2 C} = \frac{\cos^2 2C + \cos^2 C}{\sin^2 C} = \frac{(1 - 2\sin^2 C)^2 + 1 - \sin^2 C}{\sin^2 C}$

$= \frac{4\sin^4 C - 5\sin^2 C + 2}{\sin^2 C} = 4\sin^2 C + \frac{2}{\sin^2 C} - 5 \geq 2\sqrt{4\sin^2 C \cdot \frac{2}{\sin^2 C}} = 4\sqrt{2} - 5$ (当且仅当 $4\sin^2 C = \frac{2}{\sin^2 C}$ ，即 $\sin C = \frac{\sqrt{2}}{2}$ 时取等)

仅当 $4\sin^2 C = \frac{2}{\sin^2 C}$ ，即 $\sin C = \frac{\sqrt{2}}{2}$ 时取等)

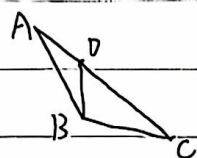
变式：由 $\tan C + \tan A = \frac{3}{2}\tan C + \tan(\frac{3}{2}\pi - 2C) = \frac{3}{2}\tan C + \frac{1}{\tan 2C} = \frac{3}{2}\tan C + \frac{1 - \tan^2 C}{2\tan C}$

$= \frac{6\tan^2 C + 1 - \tan^2 C}{2\tan C} = \frac{5\tan^2 C + 1}{2\tan C} = 5\tan C + \frac{1}{2\tan C} \geq 2\sqrt{5\tan C \cdot \frac{1}{2\tan C}} = 2\sqrt{5}$ (当且仅当

$5\tan C = \frac{1}{2\tan C}$ 则 $\tan C = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}$ 取等)

4

(8.) 由 $S_{\triangle ABC} = S_{\triangle ABD} + S_{\triangle BDC}$, 则 $\frac{1}{2}ac \sin \frac{2}{3}\pi = \frac{1}{2}BD \cdot c \sin \frac{\pi}{6} + \frac{1}{2}BD \cdot a \sin \frac{\pi}{2}$
 即 $\frac{1}{a} + \frac{2}{c} = \frac{\sqrt{3}}{3}$ 则 $a+2c = (\frac{1}{a} + \frac{2}{c}) \cdot \sqrt{3} (a+2c) = \sqrt{3} (1+4+\frac{2c}{a}+\frac{2a}{c})$
 $= 5\sqrt{3} + 2\sqrt{3}(\frac{c}{a} + \frac{a}{c}) \geq 5\sqrt{3} + 2\sqrt{3} \cdot 2\sqrt{\frac{c}{a} \cdot \frac{a}{c}} = 9\sqrt{3}$ (当且仅当 $a=c=3\sqrt{3}$ 时等号成立)



(9.) 由 $\cos C = \frac{a^2+b^2-c^2}{2ab}$, 则 $2 \cdot \frac{a^2+b^2-c^2}{2ab} = \frac{3b^2-a^2}{ab}$ 即 $b^2 = a^2 - \frac{1}{2}c^2$
 又由 $\triangle ABC$ 为锐角三角形, 则 $\begin{cases} 0 < A < \frac{\pi}{2} \\ 0 < B < \frac{\pi}{2} \\ 0 < C < \frac{\pi}{2} \end{cases}$ 则 $\begin{cases} \cos A > 0 \\ \cos B > 0 \\ \cos C > 0 \end{cases} \Rightarrow \begin{cases} b^2+c^2-a^2 > 0 \\ a^2+c^2-b^2 > 0 \\ a^2+b^2-c^2 > 0 \end{cases}$ 即 $\frac{a^2}{c^2} > \frac{3}{4}$
 即 $\frac{c^2}{a^2} < \frac{4}{3}$ 由 $\cos B = \frac{a^2+c^2-b^2}{2ac} = \frac{a^2+c^2-(a^2-\frac{1}{2}c^2)}{2ac} = \frac{3c}{4a}$
 由 $\frac{c}{a} < \frac{2\sqrt{3}}{3}$ 则 $\cos B \in (0, \frac{\sqrt{3}}{2})$ 即 $B \in (\frac{\pi}{6}, \frac{\pi}{2})$

二. (1) 边化角: 由 $b-a = 2b \sin^2 \frac{C}{2} = 2b \cdot \frac{1-\cos C}{2} = b(1-\cos C)$ 则 $b-a = b(1-\cos C)$
 即 $\sin B - \sin A = \sin B - \sin B \cos C$ 即 $\sin A = \sin B \cos C$ 又由 $\sin A = \sin(B+C) = \sin B \cos C + \sin C \cos B$
 即 $\sin C \cos B = 0$ 又由 $C \in (0, \pi)$, 则 $\sin C \neq 0$ 即 $B = \frac{\pi}{2}$ 即 $\triangle ABC$ 为直角三角形
 角化边: 由上面, 则 $b-a = b - b \cos C$ 又由 $\cos C = \frac{b^2+a^2-c^2}{2ab}$, 即 $b-a = b - \frac{b^2+a^2-c^2}{2a}$
 则 $2a(b-a) = 2ab - (b^2+a^2-c^2)$ 即 $a^2+c^2 = b^2$ 即 $\triangle ABC$ 为直角三角形

(2) 边化角: 由 $a \cos A + b \cos B = c \cos C$, 则 $\sin A \cos A + \sin B \cos B = \sin C \cos C$,
 即 $\sin 2A + \sin 2B = \sin 2C = \sin(2\pi - 2A - 2B) = -\sin(2A+2B) = -\sin 2A \cos 2B - \sin 2B \cos 2A$
 即 $\sin 2A(1+\cos 2B) + \sin 2B(1+\cos 2A) = 2\sin A \cos A \cdot 2\cos^2 B + 2\sin B \cos B \cdot 2\cos^2 A = 0$
 即 $(\sin A \cos B + \sin B \cos A) \cos B \cos A = \sin C \cos B \cos A = 0$ 由 $C \in (0, \pi)$, 则 $\sin C \neq 0$.
 则 $A = \frac{\pi}{2}$ 或 $B = \frac{\pi}{2}$

角化边: 由 $a \cdot \frac{b^2+c^2-a^2}{2bc} + b \cdot \frac{a^2+c^2-b^2}{2ac} = c \cdot \frac{a^2+b^2-c^2}{2ab}$ 即 $a^2(b^2+c^2-a^2) + b^2(a^2+c^2-b^2) = c^2(a^2+b^2-c^2)$
 即 $2a^2b^2 + c^2(a^2+b^2) - a^4 - b^4 = c^2(a^2+b^2) - c^4$
 则 $a^4 + b^4 - 2a^2b^2 = c^4$ 即 $(a^2-b^2)^2 = c^4$ 则 $a^2-b^2 = c^2$ 或 $a^2-b^2 = -c^2$
 即 $a^2 = b^2+c^2$ 或 $b^2 = a^2+c^2$

(3) 边化角: 由 $(a^2+b^2)\sin(A-B) = (a^2-b^2)\sin C = (a^2-b^2)\sin(A+B)$
 即 $(a^2+b^2)(\sin A \cos B - \sin B \cos A) = (a^2-b^2)(\sin A \cos B + \sin B \cos A)$
 则 $\sin A \cos B [(a^2+b^2) - (a^2-b^2)] = \sin B \cos A [(a^2-b^2) + (a^2+b^2)]$
 即 $\sin A \cos B \cdot b^2 = \sin B \cos A \cdot a^2$ 则 $\sin A \cos B \sin^2 B = \sin B \cos A \sin^2 A$
 由 $A \in (0, \pi)$, $B \in (0, \pi)$ 即 $\sin A \neq 0$, $\sin B \neq 0$. 则 $\cos B \sin B = \cos A \sin A$
 即 $\sin 2B = \sin 2A$ 即 $2B = 2A$ 或 $2B + 2A = \pi$ 即 $A = B$ 或 $C = \frac{\pi}{2}$

(3) 角化边: 由 $(a^2+b^2)(\sin A \cos B - \sin B \cos A) = (a^2-b^2)(\sin A \cos B + \sin B \cos A)$

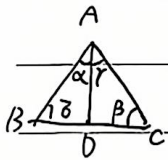
$$\text{则 } (a^2+b^2) \left(a \frac{a^2+c^2-b^2}{2ac} - b \frac{b^2+c^2-a^2}{2bc} \right) = (a^2-b^2) \left(a \frac{a^2+c^2-b^2}{2ac} + b \frac{b^2+c^2-a^2}{2bc} \right)$$

$$\text{即 } (a^2+b^2) [(a^2+c^2-b^2) - (b^2+c^2-a^2)] = (a^2-b^2) [(a^2+c^2-b^2) + (b^2+c^2-a^2)]$$

$$\text{即 } 2(a^2+b^2)(a^2-b^2) = 2(a^2-b^2) \cdot c^2 \quad \text{即 } (a^2-b^2)(a^2+b^2-c^2) = 0$$

$$\text{即 } a=b \text{ 或 } a^2+b^2=c^2$$

(4) 设 $\angle BAD = \alpha$, $\angle ACB = \beta$, 则 $\alpha + \beta = 90^\circ$ 又由 $\alpha + \beta + \gamma + \delta = 180^\circ$ 则 $\gamma + \delta = 90^\circ$



$$\text{由正弦定理: } \frac{BD}{\sin \alpha} = \frac{AD}{\sin \delta} \text{ 和 } \frac{CD}{\sin \gamma} = \frac{AD}{\sin \beta} \quad \text{则 } AD = \frac{\sin \delta}{\sin \alpha} BD = \frac{\sin \beta}{\sin \gamma} CD$$

$$\text{又由 } BD = CD \quad \text{即 } \frac{\sin \delta}{\sin \alpha} = \frac{\sin \beta}{\sin \gamma} \quad \text{则 } \sin \delta \sin \gamma = \sin \alpha \sin \beta$$

$$\text{则 } \sin \delta \cos \delta = \sin \alpha \cos \alpha \quad \text{则 } \sin 2\delta = \sin 2\alpha \quad \text{则 } \alpha = \delta \text{ 或 } \alpha + \delta = 90^\circ$$

若 $\alpha = \delta$, 则 由 $\alpha + \beta = 90^\circ$, 则 $\delta + \beta = 90^\circ$ 即 $AB \perp AC$

若 $\alpha + \delta = 90^\circ$, 则 由 $\alpha + \beta = 90^\circ$, 则 $\beta = \delta$ 即 $AB = AC$

(5) 边化角: $\sin^2 A = \sin B \cos C + \sin C \cos B = \sin A$ 即 $\sin^2 A = \sin A$

由 $A \in (0, \pi)$, $\sin A \neq 0$ 即 $\sin A = 1$ 则 $A = \frac{\pi}{2}$

$$\text{角化边: 由 } a \sin A = b \cos C + c \cos B, \text{ 则 } \sin A = \frac{b}{a} \cdot \frac{a^2+b^2-c^2}{2ab} + \frac{c}{a} \cdot \frac{a^2+c^2-b^2}{2ac} = \frac{2a^2}{2a^2} = 1$$

$$\text{即 } A = \frac{\pi}{2}$$

(6) 边化角: 由正弦定理 $a = 2R \sin A$, $b = 2R \sin B$, $c = 2R \sin C$,

$$\text{则 } \tan A = \tan B = \tan C, \text{ 即 } A = B = C$$

$$\text{角化边: } \frac{2abc}{b^2+c^2-a^2} = \frac{2abc}{a^2+c^2-b^2} = \frac{2abc}{a^2+b^2-c^2} \quad \text{即 } b^2+c^2-a^2 = a^2+c^2-b^2 = a^2+b^2-c^2$$

$$\text{即 } a^2 = b^2 = c^2 \quad \text{则 } a = b = c$$

三 完成下列求值

$$(1) \text{ 由 } \frac{a}{\sin A} = \frac{c}{\sin C}, \text{ 则 } \frac{1}{\frac{1}{2}} = \frac{\sqrt{3}}{\sin C} \quad \text{则 } \sin C = \frac{\sqrt{3}}{2} \quad \text{又由 } a < c, \text{ 则 } A < C \quad \text{则 } C = \frac{\pi}{3} \text{ 或 } \frac{2\pi}{3}$$

$$(2) \text{ 法一: 联立 } \begin{cases} b+c=3 \\ bc=2 \end{cases} \quad \text{则 } \begin{cases} b=2 \\ c=1 \end{cases} \text{ 或 } \begin{cases} b=1 \\ c=2 \end{cases} \quad \text{则 } \cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{1}{4}$$

$$\text{法二: 由 } \cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{(b+c)^2-a^2}{2bc} - 1 = \frac{3^2-2^2}{2 \times 2} - 1 = \frac{1}{4}$$

$$(3) \text{ 法一: } c = 2R \sin C \quad \text{则 } 2R \sin C = 1, \text{ 又由 } a = 2R \sin A \quad \text{则 } 2R \sin A \cos B = 2 \sin C$$

$$\text{则 由 } 2R \sin C = 1 \quad \text{则 } \frac{1}{\sin C} \sin A \cos B = 2 \sin C \quad \text{则 } \sin A \cos B = 2 \sin^2 C. \text{ 又由 } \cos A \sin B = \frac{1}{2}$$

$$\text{则 } \sin C = \sin(A+B) = \sin A \cos B + \sin B \cos A = 2 \sin^2 C + \frac{1}{2} \quad \text{即 } 16 \sin^2 C - 8 \sin C + 1 = 0$$

$$\text{则 } [4 \sin C - 1]^2 = 0 \quad \text{即 } \sin C = \frac{1}{4}$$

法二: 由 $C=1$, 则 $a \cos B = 2 \sin C = 2 \times \sin C = 2c \sin C$ 则 $\sin A \cos B = 2 \sin^2 C$ 余同上

(4) 由 $A+B=3C$, $A+B+C=\pi$ 则 $C=\frac{\pi}{4}$, 又由 $\sin B = 2 \sin(A-C)$ 则 $\sin(A+C) = 2 \sin(A-C)$

$$\text{则 } \sin A \cos C + \sin C \cos A = 2(\sin A \cos C - \sin C \cos A) \quad \text{则 } \sin A \cos C = 3 \sin C \cos A$$

$$\text{由 } \sin C = \cos C = \frac{\sqrt{2}}{2} \quad \text{则 } \sin A = 3 \cos A \quad \text{又由 } \sin^2 A + \cos^2 A = 1 \quad \text{则 } \sin A = \frac{3}{\sqrt{10}} = \frac{3}{10} \sqrt{10}$$

$$(5) \text{ 由 } a \cos B = \frac{3}{5}, b \cos A = \frac{2}{5}, \text{ 则由 } \frac{\sin(A-B)}{\sin C} = \frac{\sin(A-B)}{\sin(A+B)} = \frac{\sin A \cos B - \sin B \cos A}{\sin A \cos B + \sin B \cos A} = \frac{a \cos B - b \cos A}{a \cos B + b \cos A}$$

$$= \frac{\frac{3}{5} - \frac{2}{5}}{\frac{3}{5} + \frac{2}{5}} = \frac{1}{5} = \frac{1}{5}$$

$$(6) \text{ 由 } \sin(B-C) \tan A = \sin B \sin C \quad \text{则 } \sin(B-C) \frac{\sin A}{\cos A} = \sin B \sin C \quad \text{则 } \sin(B-C) \sin A = \cos A \sin B \sin C$$

$$\text{由 } \sin(B-C) \sin A = \sin(B-C) \sin(B+C) = (\sin B \cos C - \sin C \cos B)(\sin B \cos C + \sin C \cos B) = \sin^2 B \cos^2 C - \sin^2 C \cos^2 B$$

$$= \sin^2 B (1 - \sin^2 C) - \sin^2 C (1 - \sin^2 B) = \sin^2 B - \sin^2 C \quad \text{即 } \sin^2 B - \sin^2 C = \sin B \sin C \cos A$$

$$\text{则 } b^2 - c^2 = bc \cdot \frac{b^2 + c^2 - a^2}{2bc} \quad \text{即 } b^2 + a^2 = 3c^2 \quad \text{则 } \frac{a^2 + b^2}{c^2} = 3$$

$$(7) \text{ 由 } a^2 + b^2 = 3c^2, \text{ 则 } \frac{1}{\tan A} + \frac{1}{\tan B} - \frac{1}{\tan C} = \frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} - \frac{\cos C}{\sin C} = \frac{\cos A \sin B + \sin A \cos B}{\sin A \sin B}$$

$$- \frac{\cos C}{\sin C} = \frac{\sin C}{\sin A \sin B} - \frac{\cos C}{\sin C} = \frac{\sin^2 C - \sin A \sin B \cos C}{\sin A \sin B \sin C} = \frac{c^2 - ab \cdot \frac{a^2 + b^2 - c^2}{2ab}}{ab \sin C} = \frac{3c^2 - a^2 - b^2}{2ab \sin C} = 0$$

$$(8) \text{ 由 } a + c = 2b, \text{ 则 } \sin A + \sin C = 2 \sin B \quad \text{由 } A = \frac{A+C}{2} + \frac{A-C}{2}, C = \frac{A+C}{2} - \frac{A-C}{2}$$

$$\text{则 } \sin\left(\frac{A+C}{2} + \frac{A-C}{2}\right) + \sin\left(\frac{A+C}{2} - \frac{A-C}{2}\right) = 2 \sin \frac{A+C}{2} \cos \frac{A-C}{2} \quad \text{又由 } \sin B = \sin(A+C) = 2 \sin \frac{A+C}{2} \cos \frac{A+C}{2}$$

$$\text{由 } \frac{A+C}{2} \in (0, \frac{\pi}{2}) \quad \text{则 } \sin \frac{A+C}{2} \neq 0 \quad \text{则 } \cos \frac{A-C}{2} = 2 \cos \frac{A+C}{2} \quad \text{则 } \cos \frac{A}{2} \cos \frac{C}{2} + \sin \frac{A}{2} \sin \frac{C}{2} =$$

$$2 \cos \frac{A}{2} \cos \frac{C}{2} - 2 \sin \frac{A}{2} \sin \frac{C}{2} \quad \text{则 } 3 \sin \frac{A}{2} \sin \frac{C}{2} = \cos \frac{A}{2} \cos \frac{C}{2} \quad \text{则 } \tan \frac{A}{2} \tan \frac{C}{2} = \frac{1}{3}$$

$$(9) \text{ 由 } a \cos B - b \cos A = \frac{3}{5}c, \text{ 则 } \sin A \cos B - \sin B \cos A = \frac{3}{5} \sin C = \frac{3}{5}(\sin A \cos B + \sin B \cos A)$$

$$\text{则 } \frac{2}{5} \sin A \cos B = \frac{3}{5} \sin B \cos A \quad \text{即 } 4 \sin B \cos A = \sin A \cos B \quad \text{则 } \frac{\tan A}{\tan B} = \frac{\sin A \cos B}{\sin B \cos A} = 4$$

$$(10) \text{ 由 } \frac{b}{a} + \frac{a}{b} = \frac{a^2 + b^2}{ab}, \text{ 又由 } \cos C = \frac{a^2 + b^2 - c^2}{2ab} \quad \text{则 } \frac{a^2 + b^2}{ab} = 3 \frac{a^2 + b^2 - c^2}{ab} \quad \text{则 } 2a^2 + 2b^2 = 3c^2$$

$$\text{即 } a^2 + b^2 = \frac{3}{2}c^2 \quad \text{由 } \frac{\tan C}{\tan A} + \frac{\tan C}{\tan B} = \frac{\cos A \sin C}{\sin A \cos C} + \frac{\cos B \sin C}{\sin B \cos C} = \frac{(\cos A \sin B + \sin A \cos B) \sin C}{\sin A \sin B \cos C}$$

$$= \frac{\sin^2 C}{\sin A \sin B \cos C} = \frac{c^2}{ab \cdot \frac{a^2 + b^2 - c^2}{2ab}} = \frac{2c^2}{a^2 + b^2 - c^2} = \frac{2c^2}{\frac{3}{2}c^2 - c^2} = 4$$

$$(11) \text{ 由 } C = \frac{\pi}{3}, \sqrt{7}a = 3c, \text{ 则 } \sqrt{7} \sin A = 3 \sin C \quad \text{即 } \sin A = \frac{3\sqrt{3}}{2\sqrt{7}} = \frac{3\sqrt{21}}{14}$$

$$\text{由 } \frac{1}{\tan A} + \frac{1}{\tan B} = \frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} = \frac{\cos B \sin A + \sin B \cos A}{\sin A \sin B} = \frac{\sin C}{\sin A \sin B} = \frac{c}{a \sin B}$$

$$\text{又由 } \frac{c}{a} = \frac{\sqrt{7}}{3}, \sin B = \sin\left(\frac{\pi}{3} + A\right) \quad \text{由 } \triangle ABC \text{ 为锐角三角形, 则 } \cos A = \frac{1}{2\sqrt{7}} = \frac{\sqrt{7}}{14}$$

$$\text{则 } \sin B = \sin \frac{\pi}{3} \cos A + \cos \frac{\pi}{3} \sin A = \frac{\sqrt{3}}{2} \times \frac{\sqrt{7}}{14} + \frac{1}{2} \times \frac{3\sqrt{21}}{14} = \frac{4\sqrt{3}}{28} = \frac{\sqrt{3}}{7}$$

$$\text{则 } \frac{1}{\tan A} + \frac{1}{\tan B} = \frac{c}{a \sin B} = \frac{\sqrt{7}}{3} \times \frac{7}{\sqrt{21}} = \frac{7}{3}\sqrt{3}$$

$$(12) \text{ 由 } \sin C = 2 \sin B \cos B \quad \text{则 } \sin C = \sin 2B \quad \text{则 } 2B = C \text{ 或 } 2B + C = \pi. \text{ 由 } C = \frac{2}{3}\pi$$

$$\text{则 } B = \frac{\pi}{3} (\text{舍}) \text{ 或 } \frac{\pi}{6} \quad \text{则 } B = \frac{\pi}{6}$$

$$(13) \text{ 由 } \sin A \sin \frac{A+C}{2} = \sin B \sin A, \text{ 且 } \sin A \neq 0 \quad \text{则 } \sin \frac{A+C}{2} = \sin B = 2 \sin \frac{A+C}{2} \cos \frac{A+C}{2}, \text{ 由 } \sin \frac{A+C}{2} > 0 \quad \text{则 } \cos \frac{A+C}{2} = \frac{1}{2}$$

$$\text{则 } \frac{A+C}{2} = \frac{\pi}{3} \quad \text{则 } A+C = \frac{2}{3}\pi \quad \text{又由 } A+B+C=\pi \quad \text{则 } B = \frac{\pi}{3}$$

四. 综合题

1.

$$\text{由 } \vec{CD} = \frac{2}{3}\vec{CB} + \frac{1}{3}\vec{CA}, \text{ 则 } \vec{CD} = \frac{1}{9}(4\vec{CB}^2 + \vec{CA}^2 + 4\vec{CB} \cdot \vec{CA})$$

$$\text{则 } 4 = \frac{1}{9}(4a^2 + b^2 - 2ab) \quad \text{则 } 4a^2 + b^2 - 2ab = 36 \quad ①$$

$$\text{由余弦定理: } AB^2 = a^2 + b^2 - 2ab \cos C = a^2 + b^2 + ab, \text{ 又由 } AB = 6$$

$$\text{则 } 36 = a^2 + b^2 + ab \quad ② \quad \text{由 } ① - ② \text{ 得, } a = b, \text{ 代入 } ②, \text{ 得 } a = b = 2\sqrt{3}$$

$$\therefore \text{则 } S_{\triangle ABC} = \frac{1}{2}ab \sin C = \frac{1}{2} \times (2\sqrt{3})^2 \times \frac{\sqrt{3}}{2} = 3\sqrt{3}$$

2. 由 $\frac{a - \cos B}{a - \cos C} = \frac{b}{c}$ 则 $ac - c \cos B = ab - b \cos C$ 则 $ac - ab = c \cos B - b \cos C$

$$\text{则 } ac - ab = c \cdot \frac{a^2 + c^2 - b^2}{2ac} - b \cdot \frac{a^2 + b^2 - c^2}{2ab} \quad \text{则 } a^2(c - b) = c^2 - b^2 \quad \text{又由 } b \neq c, \text{ 则 } a^2 = c + b$$

3. 由 $BD \sin \angle ABC = a \sin C$ 则 $BD \cdot b = ac$, 又由 $b^2 = ac$, 则 $BD = b$

$$\text{由 } \vec{BD} = \frac{2}{3}\vec{BC} + \frac{1}{3}\vec{BA}, \text{ 则 } |\vec{BD}|^2 = \frac{1}{9}(4\vec{BC}^2 + \vec{BA}^2 + 4|\vec{BC}||\vec{BA}|\cos \angle ABC) \quad \text{则}$$

$$b^2 = \frac{1}{9}(4a^2 + c^2 + 4ac \cos \angle ABC) \quad \text{即 } 9ac = 4a^2 + c^2 + 4ac \cos \angle ABC \quad ①$$

$$\text{又由余弦定理: } b^2 = a^2 + c^2 - 2ac \cos \angle ABC \quad \text{则 } ac = a^2 + c^2 - 2ac \cos \angle ABC \quad ②$$

$$\text{由 } ① + 2 \times ② \text{ 得, } 11ac = 6a^2 + 3c^2 \quad \text{则 } 6a^2 - 11ac + 3c^2 = 0 \quad \text{则 } (2a - 3c)(3a - c) = 0$$

$$\text{即 } a = \frac{3}{2}c \text{ 或 } a = \frac{1}{3}c \quad \text{则代入 } ②, \text{ 得 } \cos \angle ABC = \frac{7}{6} \text{ (舍) 或 } \frac{7}{12}$$

4. 由 $a^2 = b^2 + bc$, 又由 $a^2 = b^2 + c^2 - 2bc \cos A$ 则 $c^2 - 2bc \cos A = bc$ 则 $c - 2b \cos A = b$

$$\text{则 } c = b + 2b \cos A \quad \text{则 } \sin C = \sin B + 2 \sin B \cos A \quad \text{又由 } \sin C = \sin B \cos A + \sin A \cos B$$

$$\text{则 } \sin B = \sin C - 2 \sin B \cos A = \sin A \cos B - \sin B \cos A = \sin(A - B) \quad \text{则 } B = A - B \text{ 即 } A = 2B$$

5. 由 $\frac{\sin 2B}{1 + \cos 2B} = \frac{2 \sin B \cos B}{2 \cos^2 B} = \frac{\sin B}{\cos B}$, 又由 $\frac{\sin 2B}{1 + \cos 2B} = \frac{\cos A}{1 + \sin A}$ 则 $\frac{\sin B}{\cos B} = \frac{\cos A}{1 + \sin A}$

$$\text{则 } \sin B + \sin B \sin A = \cos A \cos B \quad \text{则 } \sin B = \cos A \cos B - \sin A \sin B = \cos(A + B), \text{ 又由 } \sin B = \cos(\frac{\pi}{2} - B)$$

$$\text{则 } \cos(\frac{\pi}{2} - B) = \cos(A + B) \quad \text{则 } \frac{\pi}{2} - B = A + B \quad \text{又由 } A + B = \pi - C, \text{ 则 } \frac{\pi}{2} - B = \pi - C$$

$$\text{即 } C = B + \frac{\pi}{2}$$

6. 由 $1 + \sin 2C = (\sin C + \cos C)^2$, 则 $\frac{b^2}{a^2} = (\sin C + \cos C)^2$ 又由 $\sin C + \cos C = \sqrt{2} \sin(C + \frac{\pi}{4})$

$$\text{由 } C \in (0, \frac{3}{4}\pi), \text{ 则 } C + \frac{\pi}{4} \in (\frac{\pi}{4}, \pi) \quad \text{则 } \sin(C + \frac{\pi}{4}) > 0 \quad \text{则 } \frac{b}{a} = \sin C + \cos C \quad \text{则 } \frac{\sin B}{\sin A} = \sin C + \cos C$$

$$\text{则 } \sin B = \sin A \sin C + \sin A \cos C \quad \text{又由 } \sin B = \sin A \cos C + \sin C \cos A \quad \text{则 } \sin C \cos A = \sin A \sin C$$

$$\text{又由 } C \in (0, \frac{3}{4}\pi), \text{ 则 } \sin C \neq 0 \quad \text{则 } \sin A = \cos A \quad \text{则 } \tan A = 1 \quad \text{由 } A \in (0, \pi), \text{ 则 } A = \frac{\pi}{4}$$

7. 由 $\frac{\sin C}{\sin B} = \frac{2\sqrt{2}}{3}$ 则 $\sin C = \frac{2\sqrt{2}}{3} \sin B$ 又由 $\sin C = \sin(B + A) = \sin B \cos A + \sin A \cos B =$

$$\frac{\sqrt{2}}{2} \sin B + \frac{\sqrt{2}}{2} \cos B \quad \text{则 } \frac{\sqrt{2}}{2} \cos B + \frac{\sqrt{2}}{2} \sin B = \frac{2\sqrt{2}}{3} \sin B \quad \text{则 } \frac{\sqrt{2}}{2} \cos B = \frac{\sqrt{2}}{6} \sin B \quad \text{即 } \sin B = 3 \cos B$$

$$\text{又由 } \sin^2 B + \cos^2 B = 1, \text{ 解得 } \sin B = \frac{3}{\sqrt{10}}, \cos B = \frac{1}{\sqrt{10}} \quad \text{则 } S_{\triangle ABC} = \frac{1}{2}ac \sin B = \frac{1}{2} \times 2\sqrt{6} \times \frac{3}{\sqrt{10}} = 3$$

8. 由 $A=135^\circ$, $\angle B=\angle D=90^\circ$, 则 $\angle C=45^\circ$. 由 $AB=2$, $AD=\sqrt{2}$, 则由余弦定理:
 $BD^2 = AD^2 + AB^2 - 2AD \cdot AB \cos A = 10$, 则 $BD=\sqrt{10}$. 又由正弦定理 $\frac{BD}{\sin A} = 2R$
 则 $\triangle ABD$ 外接圆直径为 $2\sqrt{5}$ 又由 $\angle B=\angle D=90^\circ$, 则 A, B, C, D 四点共圆.
 则由 $\angle B=\angle D=90^\circ$, 则 AC 为直径, $AC=2\sqrt{5}$. 由勾股定理,
 则 $BC = \sqrt{AC^2 - AB^2} = \sqrt{20 - 4} = \sqrt{16} = 4$, $CD = \sqrt{AC^2 - AD^2} = \sqrt{18} = 3\sqrt{2}$
 则 $S_{\text{四边形}ABCD} = S_{\triangle ABC} + S_{\triangle ADC} = \frac{1}{2}AB \cdot BC + \frac{1}{2}AD \cdot CD = \frac{1}{2} \times 2 \times 4 + \frac{1}{2} \times \sqrt{2} \times 3\sqrt{2} = 7$

9. 由 $c^2 \sin A = 6 \sin C$, 则 $c^2 \cdot a = 6c$ 则 $ac = 6$ 由 $\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{(ac)^2 - b^2}{2ac} - 1$
 $= \frac{18}{2 \times 6} - 1 = \frac{1}{2}$ 由 $B \in (0, \pi)$, 则 $B = \frac{\pi}{3}$ 则 $\sin B = \frac{\sqrt{3}}{2}$ 即 $S_{\triangle ABC} = \frac{1}{2}ac \sin B = \frac{1}{2} \times 6 \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$

10. 由圆内接四边形对顶角互补, 则 $\angle BAD = 120^\circ$, 由余弦定理:

$BD^2 = 5^2 + 3^2 - 2 \times 3 \times 5 \times (-\frac{1}{2}) = 49$ 则 $BD = 7$
 由余弦定理: $BD^2 = BC^2 + 5^2 - 2 \times 5 \times BC \times \frac{1}{2}$ 则 $BC^2 - 5BC - 24 = 0$

则 $(BC - 8)(BC + 3) = 0$ 则 $BC = 8$

11. 由 $\cos(\alpha - \beta) = 2 \sin(\alpha + \frac{\pi}{3}) \sin(\beta + \frac{\pi}{3}) = 2(\frac{1}{2} \sin \alpha + \frac{\sqrt{3}}{2} \cos \alpha)(\frac{1}{2} \sin \beta + \frac{\sqrt{3}}{2} \cos \beta)$

又由 $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$ 则 $\cos \alpha \cos \beta + \sin \alpha \sin \beta = \frac{1}{2} \sin \alpha \sin \beta +$

$\frac{3}{2} \cos \alpha \cos \beta + \frac{\sqrt{3}}{2} (\sin \beta \cos \alpha + \sin \alpha \cos \beta)$ 即 $\frac{1}{2} \sin \alpha \sin \beta - \frac{1}{2} \cos \alpha \cos \beta =$

$-\frac{1}{2} (\cos \alpha \cos \beta - \sin \alpha \sin \beta) = -\frac{1}{2} \cos(\alpha + \beta) = \frac{\sqrt{3}}{2} \sin(\alpha + \beta)$

则 $\sqrt{3} \sin(\alpha + \beta) = -\cos(\alpha + \beta)$ 则 $\tan(\alpha + \beta) = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$ 由 $\alpha + \beta \in (0, \pi)$

则 $\alpha + \beta = \frac{5\pi}{6}$ 由 $\alpha + \beta + \angle ADB = \pi$, 则 $\angle ADB = \frac{\pi}{6}$

12. 由 $\sin A \sin B + \sin B \sin C = 1 - \cos 2B = 2 \sin^2 B$ 则 $ab + bc = 2b^2$ 则 $2b = a + c$

由 $c^2 = a^2 + b^2 - 2ab \cos C = a^2 + b^2 + \sqrt{2}ab$ 又由 $c = 2b - a$ 则 $(2b - a)^2 = a^2 + b^2 + \sqrt{2}ab$

则 $3b = (4 + \sqrt{2})a$ 则 $\frac{\sin A}{\sin B} = \frac{a}{b} = \frac{3}{4 + \sqrt{2}} = \frac{12 - 3\sqrt{2}}{14}$

13 由 $\angle BAD + \angle BCD = \pi$, 则 A, B, C, D 四点共圆.

设 $\angle BAD = \alpha$ 则由余弦定理: $BD^2 = 2^2 + 4^2 - 2 \times 2 \times 4 \times \cos \alpha = 20 - 16 \cos \alpha$

由圆周角定理, 则 $\angle BAC = \angle BDC$, 且 $\angle BAC = \frac{\alpha}{2}$ 则 $\angle BDC = \frac{\alpha}{2}$

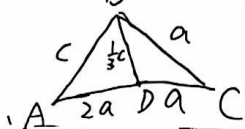
由正弦定理: $\frac{BC}{\sin \frac{\alpha}{2}} = \frac{BD}{\sin \angle BCD} = \frac{BD}{\sin(\pi - \alpha)} = \frac{BD}{\sin \alpha}$ 同理, $\angle CBD = \frac{\alpha}{2}$.

则 $\frac{CD}{\sin \frac{\alpha}{2}} = \frac{BD}{\sin \alpha}$ 则 $BD = \frac{BC \sin \alpha}{\sin \frac{\alpha}{2}} = 2 \cos \frac{\alpha}{2} BC$, $BD = 2 \cos \frac{\alpha}{2} CD$.

由 $\frac{BD^4}{BC \cdot CD} = \frac{BD^2}{BC \cdot CD} \cdot BD^2 = \frac{2 \cos \frac{\alpha}{2} \cdot CD \cdot 2 \cos \frac{\alpha}{2} BC}{BC \cdot CD} \cdot (20 - 16 \cos \alpha) = 4 \cos^2 \frac{\alpha}{2} \times 4(5 - 4 \cos \alpha) =$

$16 \cos^2 \frac{\alpha}{2} [5 - 4(2 \cos^2 \frac{\alpha}{2} - 1)] = 16 \cos^2 \frac{\alpha}{2} (9 - 8 \cos^2 \frac{\alpha}{2}) = 2 \times 8 \cos^2 \frac{\alpha}{2} (9 - 8 \cos^2 \frac{\alpha}{2}) \leq 2 \cdot (\frac{8 \cos^2 \frac{\alpha}{2} + 9 - 8 \cos^2 \frac{\alpha}{2}}{2})^2$
 $= \frac{81}{2}$ (当且仅当 $8 \cos^2 \frac{\alpha}{2} = 9 - 8 \cos^2 \frac{\alpha}{2}$ 时取得等号, 则 $\cos \frac{\alpha}{2} = \frac{3}{4}$ 时取得)

14. B



由 $AC = 3BC$, 则 $b = 3a$, 又由 $\vec{AD} = 2\vec{DC}$, 则 $AD = 2a$, $CD = a$

由 $\sin \angle BDC = 3\sin \angle BAC$, 则 $\sin \angle BDC = 3\sin A$. 由 $\angle BDC + \angle BDA = \pi$

则 $3\sin A = \sin \angle BDA$ 又由 $\frac{BD}{\sin A} = \frac{AB}{\sin \angle BDA}$ 则 $BD = \frac{AB \sin A}{\sin \angle BDA} = \frac{1}{3}c$

由 $\angle BDA + \angle BDC = \pi$ 则 $(2a)^2 + (\frac{1}{3}c)^2 - c^2 = -\frac{a^2 + (\frac{1}{3}c)^2 - a^2}{2 \times \frac{1}{3}c}$ 即 $c^2 = 6a^2$

则 $c = \sqrt{6}a$ 由 $\cos \angle ACB = \frac{(3a)^2 + a^2 - c^2}{2 \times 3a \times a} = \frac{10a^2 - 6a^2}{6a^2} = \frac{2}{3}$ 则 $|\vec{CA} \cdot \vec{CB} - |\vec{AB}|^2| =$

$3a \cdot a \times \frac{2}{3} - c = 2a^2 - \sqrt{6}a$ 易知, 当 $a = -\frac{\sqrt{6}}{2 \times 2} = -\frac{\sqrt{6}}{4}$ 时, 取得最小值.

则 $\cos \angle ACB = \frac{2}{3}$ 又由 $\sin^2 \angle ACB + \cos^2 \angle ACB = 1$. 则 $\sin \angle ACB = \frac{\sqrt{5}}{3}$.

则 $S_{\triangle ABC} = \frac{1}{2} \cdot 3a \cdot a \cdot \frac{\sqrt{5}}{3} = \frac{3}{16}\sqrt{5}$

15: (1) 由 $BD^2 = AB^2 + AD^2$ 则 $BD = 2$ 又由 $CD = 2$, 则 $\cos \angle BDC = \frac{2^2 + 2^2 - \sqrt{2}^2}{2 \times 2 \times 2} = \frac{3}{4}$

又由 $\cos \angle BDC > 0$. 则 $\angle BDC$ 为锐角. 则 $\sin \angle BDC = \frac{\sqrt{7}}{4}$ 又由 $\sin \angle APB = \frac{1}{2}$,

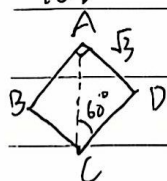
$\cos \angle APB = \frac{\sqrt{3}}{2}$ 则 $\sin \angle ADC = \sin(\angle ADB + \angle BDC) = \frac{1}{2} \times \frac{3}{4} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{7}}{4} = \frac{\sqrt{2} + 3}{8}$

(2) 由 $C = \frac{\pi}{4}$, $BD = 2$. 则 $BD^2 = BC^2 + CD^2 - 2BC \cdot CD \cos C$ 则 $CD^2 - 2CD - 2 = 0$

则 $CD = \sqrt{2} + 1$ 或 $-\sqrt{2} + 1$ (舍) 则 $S_{\text{四边形}ABCD} = S_{\triangle ABD} + S_{\triangle BCD} = \frac{1}{2} \times 1 \times \sqrt{2} + \frac{1}{2} \times \sqrt{2}(\sqrt{2} + 1) \times \frac{\sqrt{2}}{2}$

$= \frac{2\sqrt{2} + 1}{2} = \sqrt{2} + \frac{1}{2}$

16.



(1) 由 $2S = -\sqrt{3} \vec{BA} \cdot \vec{BC}$ 则 $S = -\frac{\sqrt{3}}{2} \vec{BA} \cdot \vec{BC} = -\frac{\sqrt{3}}{2} |\vec{BA}| \cdot |\vec{BC}| \cos B$

又由 $S = \frac{1}{2} |\vec{BA}| |\vec{BC}| \sin B$ 则 $-\frac{\sqrt{3}}{2} \cos B = \frac{1}{2} \sin B$ 则 $\tan B = -\sqrt{3}$

由 $B \in (0, \pi)$ 则 $B = \frac{2}{3}\pi$.

(2) 又由 $\cos \angle ADC = \frac{1}{2}$, $\angle ADC \in (0, \pi)$, 则 $\angle ADC = \frac{\pi}{3}$.

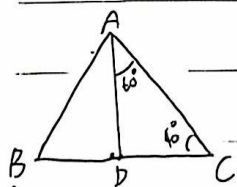
则 $\triangle ADC$ 是等边三角形. 则 $CD = \sqrt{3}$. 设 $AB = t$. 则 $\angle A \perp AD$

则 $\angle BAD = \frac{\pi}{2}$ 又由 $\angle DAC = \frac{\pi}{3}$ 则 $\angle BAC = \frac{\pi}{6}$ 又由 $\angle B = \frac{2}{3}\pi$ 则 $\frac{AC}{\sin B} = \frac{BC}{\sin \angle BAC}$

则 $BC = 1$ 又由 $\angle BAC + \angle B + \angle BCA = \pi$ 则 $\angle BCA = \frac{\pi}{6}$ 由 $\angle BAC = \angle BCA = \frac{\pi}{6}$,

则 $AB = BC = 1$. 则 四边形 $ABCD$ 周长 $= 2\sqrt{3} + 2$

17.



由 $BD = AC$, 则 由正弦定理: $\frac{BD}{\sin \angle BAD} = \frac{AD}{\sin \angle ABD}$, 又由 $\frac{AC}{\sin \angle ADC} = \frac{AD}{\sin \angle ACD}$

则 $\frac{\sin \angle ABD}{\sin \angle BAD} = \frac{AD}{BD} = \frac{AD}{AC} = \frac{\sin \angle ACD}{\sin \angle ADC}$ 则 设 $\angle ABD = \theta$

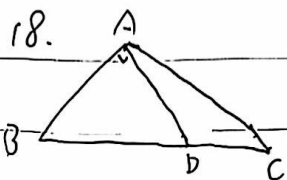
由外角和, 则 $\angle BDA = 100^\circ$ 则 $\frac{\sin \theta}{\sin(180^\circ - \theta)} = \frac{\sin 40^\circ}{\sin 80^\circ} = \frac{1}{2 \cos 40^\circ}$

则 $2 \cos 40^\circ \sin \theta = \sin(80^\circ - \theta) = \sin 80^\circ \cos \theta - \cos 80^\circ \sin \theta$

则 $\sin \theta (2 \cos 40^\circ + \cos 80^\circ) = \sin 80^\circ \cos \theta$ 则 $\tan \theta = \frac{\sin 80^\circ}{2 \cos 40^\circ + \cos 80^\circ} =$

$\frac{\sin 80^\circ}{2 \cos(120^\circ - 80^\circ) + \cos 80^\circ} = \frac{\sin 80^\circ}{2(\cos 120^\circ \cos 80^\circ + \sin 120^\circ \sin 80^\circ) + \cos 80^\circ} = \frac{\sin 80^\circ}{\sqrt{3} \sin 80^\circ} = \frac{\sqrt{3}}{3}$ 由 $\theta \in (0, \pi)$

则 $\theta = \frac{\pi}{6}$



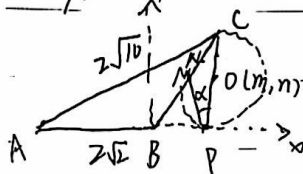
18. 由 $S_{\triangle ABC} = S_{\triangle ABD} + S_{\triangle ADC}$ 则 $\frac{1}{2} \cdot c \cdot b \cdot \frac{\sqrt{3}}{2} = \frac{1}{2} \cdot AD \cdot c + \frac{1}{2} \cdot AD \cdot b \cdot \frac{1}{2}$

则 $\sqrt{3}bc = 4c + 2b \geq 2\sqrt{4c \cdot 2b} = 4\sqrt{2bc}$ (当且仅当 $4c = 2b$

时取得等号) 则 $3b^2c^2 \geq 32bc$ 则 $bc \geq \frac{32}{3}$

又由 $S_{\triangle ABC} = \frac{1}{2}bc \sin A = \frac{\sqrt{3}}{4}bc \geq \frac{8}{3}\sqrt{3}$ ($b=2c = \frac{8}{3}\sqrt{3}$ 时取得)

19.



法一: 正弦定理. 设 $\angle CPM = \alpha$ ($\alpha \in (0, \frac{\pi}{2})$)

由 $\frac{CM}{\sin \alpha} = 2R$, 又由 $CM = 2$. 当 $\triangle CMP$ 外接圆直径最小时, 则 $\sin \alpha$ 最大.

则 α 最大 此时 $\triangle CMP$ 外接圆与直线 AB 相切.

由 $\cos \angle ABC = \frac{(2\sqrt{2})^2 + 4^2 - (2\sqrt{10})^2}{2 \times 4 \times 2\sqrt{2}} = -\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}$ 则 $\sin \angle ABC = \frac{\sqrt{2}}{2}$

建立如图所示坐标系. $B(0,0)$, $C(2\sqrt{2}, 2\sqrt{2})$, $M(\sqrt{2}, \sqrt{2})$ 设 $P(m, 0)$ ($m > 0$).

$\triangle CMP$ 外接圆心为 $O(m, n)$ ($n > 0$). 则 CM 的中垂线过圆心. 由 CM 斜率为 1

则中垂线斜率为 -1 则 CM 中垂线: $y = -(x - \frac{3}{2}\sqrt{2}) + \frac{3}{2}\sqrt{2} = -x + 3\sqrt{2}$

则 $n = -m + 3\sqrt{2}$. 由圆心到 M 距离为 R , 设 CM 中点为 N . 由点到直线距离得

则 $ON = \frac{|n-m|}{\sqrt{2}}$ 则 $OM^2 = ON^2 + MN^2$ 又由 $MN = 1$ 则 $R^2 = \frac{(m-n)^2}{2} + 1$

又由相切, 则 n 为半径 R 则 $n^2 = \frac{(m-n)^2}{2} + 1$ 又由 $n = -m + 3\sqrt{2}$

则解得 $n = \sqrt{2}$ 或 $5\sqrt{2}$ (舍) 则 $m = 2\sqrt{2}$ 或 $-2\sqrt{2}$ (舍) 则 $P(2\sqrt{2}, 0)$

由 $\vec{PM} = (-\sqrt{2}, \sqrt{2})$, $\vec{PC} = (0, 2\sqrt{2})$ 则 $\cos \alpha = \frac{\sqrt{2} \times 2\sqrt{2}}{\sqrt{2} \times \sqrt{2} \cdot 2\sqrt{2}} = \frac{\sqrt{2}}{2}$ 则 $\alpha = \frac{\pi}{4}$

法二: 函数

过 M, C 点分别作直线 AB 的垂线, 垂足分别为 N, R .

由题意, $MN = \sin \angle MBN \cdot BM$, 由 $\cos \angle ABC = -\frac{\sqrt{2}}{2}$, 则 $\angle ABC = 135^\circ$,

则 $\angle CBN = 45^\circ$. 则 $BN = \sqrt{2}$, $MN = \sqrt{2}$, $BR = 2\sqrt{2}$, $CR = 2\sqrt{2}$.

设 $BP = t$ ($t > 0$) 则 $\tan \angle MPN = \frac{MN}{NP} = \frac{\sqrt{2}}{t - \sqrt{2}}$ $\tan \angle CPR = \frac{2\sqrt{2}}{t - 2\sqrt{2}}$

则 $\tan \angle CPM = \tan (\angle CPR - \angle MPN) = \frac{\frac{2\sqrt{2}}{t - 2\sqrt{2}} - \frac{\sqrt{2}}{t - \sqrt{2}}}{1 + \frac{\sqrt{2}}{t - \sqrt{2}} \cdot \frac{2\sqrt{2}}{t - 2\sqrt{2}}} = \frac{\sqrt{2}t}{t^2 - 3\sqrt{2}t + 8} = \frac{\sqrt{2}}{t + \frac{8}{t} - 3\sqrt{2}} \leq \frac{\sqrt{2}}{2\sqrt{2} - 3\sqrt{2}}$

$= 1$ (当且仅当 $t = \frac{8}{t}$, 即 $t = 2\sqrt{2}$ 时取得等号) 此时 $\angle CPM = \frac{\pi}{4}$